

Allah University

End Semester Examination, 2022-2023

Course: Civil Engineering Year: Final Year Subject: Environmental Geo-technology

Subject Code: CENUGPE06 (PEC-VI)

Max. Marks: 80

Times: 3 hrs

Answer all Questions.

1. Write Short notes on the following:

(5x4 = 20 Marks)

- ☒ (i) Role of soil in Geoenvironmental application
- ☒ (ii) Soil-water interaction
- ☒ (iii) Components of an engineered landfill
- ☒ (iv) Specific surface area
- ☒ (v) Integrated solid waste management programme (ISWMP)

2. (a) Discuss the issues related to reuse of waste materials. $\neq$

☒ (b) Differentiate between natural attenuates and containment landfills. ✓

☒ (c) Explain the remedial methods for soil and ground water. ✓

(6+6+8 = 20 Marks)

3. ☒ (a) What are the different options for waste land filling? Discuss the advantages and disadvantages of each waste land filling.

☒ (b) What is the design life of a landfill? How the capacity of a landfill is estimated?

(10+10 = 20 Marks)

4. ☒ (a) Discuss the probable industry discharging contaminants into ground water.

☒ (b) What are the various solid waste management mechanism practiced in India. Discuss in detail.

(8+12 = 20 Marks)

Even (Spring) Semester Examination 2022-2023
Paper Code: MBAUGHU02; Paper name: Professional Values and Ethics
Civil Engineering (CEN) VIIIth Semester
Full Marks: 80; Time: 3Hrs.

(The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as possible)

GROUP: A (Answer all the questions)

(1 x 10 = 10)

1. Choose and write ONLY the correct option (a/b/c/d). DO NOT write full sentences.

I. Ethics is the science of

A. beauty

✓ B. conduct

C. truth

D. mind

II. Value is

A. response of the society

✓ B. enduring belief

C. material culture

D. non-material culture

III. The word 'ethics' was derived from the Greek word

A. ethies

✓ B. ethos

C. ethecs

D. ethise

IV. Aesthetics deals with the standard of

A. truth

✓ B. beauty

C. goodness

D. trust

V. Which country has more collectivist culture

A. USA

✓ B. Japan

C. Germany

D. China

VI. Publishing wrong information In Internet is _____

A. an issue of Morality

B. an issue of Computer ethics ✓

C. an issue of Computer equity

D. an issue of the Digital divide

VII.

Business ethics has a _____ application.

A. natural

✓ B. universal

C. practical

D. none of the above

VIII. The relevance of ethics is in its

A. Context

✓ B. Applications

C. Principles

D. Understanding

IX. Business malpractice does not include

A. Black Marketing

✓ B. Advertisement

C. Duplication

D. Adulteration

X. Dowry deaths, wife battering is an example of

A. Criminal violence

B. Domestic violence

C. Social violence

D. Gross violence

GROUP: B (Answer any five questions)

(5 x 5 = 25)

2. What do you mean by Utilitarianism?

✓ 1. What are the features of 'Rapid Technology growth'? Discuss. ✓

✓ 2. Write a short note on 'Wind Energy'. ✓

✓ 3. Write the names of five 'Environmental Regulations Acts'? ✓

✓ 4. Discuss the problems of Technology Transfers.

✓ 5. What are the problems of Man machine Interactions?

✓ 6. Write a short note on 'Canons of Ethics'. ✓

GROUP: C (Answer any three questions)

(15 x 3 = 45)

9. Discuss the Important Parameters of Environmental Ethics.

✓ 10. What is the concept of 'Club of Rome'? Discuss with suitable diagram.

✓ 11. What do you understand by 'Sustainable Development'? How the 'Sustainable Development' is linked with 'Energy Crisis & Renewable Energy Sources'? ✓

12. Justify your view on inclusion of 'Professional Values and Ethics' course in the Engineering domain.

✓ 13. Discuss the Appropriate Technology Movement of Schumacher. ✓

ALIAH UNIVERSITY

Department of Civil Engineering

B.Tech/Semester-VIII/CENUGPE05/PEC-V/2023

Repairs & Rehabilitation of Structures

Time Allotted: 03 Hours

Full Marks: 80

Candidates are requested to answer in their own words as far as practicable. The answer to one complete question, including sub-questions, must be written simultaneously.

answer any eight of the following:

(8×10 = 80)

Que. ~~1~~ What is meant by distress? Also, explain the causes and the preventive aspects of distress in structures.

Que. ~~2~~ Enlist and explain the different chemical attacks and their preventive measures of concrete.

Que. -3 What are the factors that influence the strength of concrete? Briefly discuss the effect of the w/c ratio and workability on the strength of concrete.

Que. ~~4~~ Explain the different ways of controlling the corrosion of steel in concrete. Explain any two methods.

Que. ~~5~~ Explain the Rebound hammer test conducted on concrete. List the advantages and disadvantages of using it.

Que. ~~6~~ Enlist and explain the different delamination techniques in detail.

Que. ~~7~~ Discuss the role of different types of polymer-modified concrete products for repairing jobs.

Que. ~~8~~ How do you assess the crack for repairing the job? Explain different crack repair techniques.

Que. -9 Differentiate between dry mix shotcrete and wet mix shotcrete.

Que. ~~10~~ Write short notes on any two of the following: (i) Beam shear strengthening (ii) Composite construction (iii) Passive and active structural health monitoring techniques.

Even Semester Examination 2022-23

Subject Code: CENUGPC24

Subject Name: Construction Management

B. Tech Civil Engineering: 8th Semester

Full Marks: 80 (8x10)

Time: 3 Hrs.

Attempt Any 8 Questions. Each question carries equal marks

1. Describe the techniques of planning. Discuss the importance of CPM and PERT in construction management.

2. Prepare an approximate estimate of building project with total plinth area of all building is 800sm. Use following information given below:

a) Plinth area rate Rs.450sqm

b) Cost of water supply @8% of cost of building.

c) Cost of Sanitary and Electrical installations each @ 8.5% of cost of Building.

d) Cost of architectural features @1.5% of building cost.

e) Cost of roads and lawns @5% of building cost.

Determine the total cost of the building project.

3. What are different types of project. Describe each project in detail. Also mentioned, different phases of project.

4. What are the different machineries used for construction? Write down the importance of machinery in construction.

5. A project schedule has:

Activity	1-2	1-3	2-4	3-4	3-5	4-9	5-6	5-7	6-8	7-8	8-10	9-10
Time(Day)	4	2	1	2	3	5	3	7	1	3	4	6

- ✓ a) Construct network diagram.
- ✓ b) Compute the earliest event time and latest event time.
- ✓ c) Determine the critical path and total project duration.

6. Elaborate the concept of project monitoring. What do you mean by quality control for a construction project?

✓ 7. Write down the steps involve in solving PERT network. For a given activity, the optimistic time, pessimistic time and most probable estimates are 6,18, and 9 days respectively. Determine the expected time.

✓ 8. Write down the advantages of network analysis and disadvantage of bar-chart for a project management.

✓ 9. Define the following terms:

- ✓ a) Optimistic time event
- ✓ b) Pessimistic time event
- ✓ c) Most likely time

✓ 10. The time estimates obtain from three contractors A, B and C for executing a particular job as under:

Contractor	Optimistic Time	Most likely time	Pessimistic time
A	6	11	14
B	4	10	12
C	5	9	13

Which one of these contractors are certain about completing the job on time?